

Project Scope

In Scope [OK]

- * Disease library (?25 entries) with standardized content and references.
- * Search and filters for discovery.
- * EN/RU locale support.
- * Sources/references page with provenance (source + last updated).
- * Optional research summaries/cards (read-only).
- * Platform: React frontend + Express/Prisma backend + PostgreSQL + Docker + OpenAPI.

Out of Scope [X]

Feature	Reason	When Possible
Medical diagnosis/treatment advice, t...	Medical decision-making is not the go...	Future phase / Never
User accounts for readers (bookmarks,...	Requires identity, moderation, and da...	Future phase
Advanced BI stack (Power BI/Fabric), ...	Adds complexity and infrastructure co...	Future phase
ML risk prediction or personalised re...	Requires model development, validatio...	Future phase
Native mobile apps (iOS/Android) and ...	Requires separate platform build and ...	Future phase
Paid/proprietary datasets or manual d...	Conflicts with budget constraint and ...	Never

Assumptions

#	Assumption	Impact if Wrong	Probability
1	Data availability & licensing: official medical data ingestion scope would be reduced	Reduced data ingestion scope would be reduced	Medium
2	Stability of APIs/files: data format changes	Additional work to adapt ETL/pipeline	Medium
3	No personal/medical data (PII/PSI) scope and compliance requirements	Scope and compliance requirements increase	High
4	Hosting & tooling: a standard web application	Deployment effort increases; environment complexity	Medium
5	Review access: a clinical consultation	Reduced content quality assurance	Low
6	Time allocation: consistent individual contributions	Deliverables may be reduced or delayed	Medium

Constraints

Limitations that affect the project:

Constraint Type	Description	Mitigation
Time	Timeline: ~4 months total.	Prioritize MVP features; timebox anal...
Resources	Team: single developer covering FE/BE...	Use proven libraries/frameworks; keep...
Technology	Technology stack fixed: React, Node.j...	Stick to stack defaults; avoid experi...
Budget	No paid tools/datasets; rely on open-...	Use official open datasets; avoid pai...
External/Compliance	Respect dataset licenses; GDPR-aware ...	Keep tracking optional/minimal; docum...
Academic	Deliverables must meet thesis format ...	Maintain documentation as part of dev...

Dependencies

Dependency	Type	Owner	Status	Why it matters	Management
Official datasets (WHO, Eurostat, CDC, etc.)	External	WHO/Eurostat/CDC/...	[OK]	Indicators and evidence for analysis	Track sources & licenses; mirror samp...
Python ETL toolchain (Pandas, etc.)	Technical	Python ecosystem	[OK]	Cleaning/aggregation of raw data	Pin versions; keep a small test datas...
PostgreSQL	External	PostgreSQL community	[OK]	Persistent disease counts storage	Use managed DB; migrations; backup ...
Node.js + Express	Technical	Node/Express maintainers	[OK]	API delivery for the frontend	Pin Node version; keep OpenAPI contr...
React + Vite	Technical	Meta / Vite community	[OK]	UI rendering, navigation	Pin dependencies; basic E2E smoke; ac...
React Router	Technical	React Router maintainers	[OK]	Client-side routing	Keep route constants centralized; smo...
Redux Toolkit + React Redux	Technical	Redux maintainers	[OK]	Shared client state (Redux Toolkit / Redux	Keep slices/thunks scoped by feature;...
Axios	Technical	Axios maintainers	[OK]	HTTP client for REST API	Pin base URL/interceptors; han...
Sentry (frontend)	External	Sentry	[OK]	Error tracking and performance mon...	Configure SENTRY_DSN; disable in c...
Prisma (ORM) + @prisma/client	Technical	Prisma	[OK]	Type-safe DB access layer	Version pinning; generate after s...
Redis (via ioredis)	External	Redis community	[OK]	Cache for read-heavy analytics	Test Redis caching; define key convent...
OpenAPI docs (express-openapi-swagger)	Technical	Library maintainers	[OK]	API contract documentation	Keep docs close to code; verify /api-...
Inversify (DI)	Technical	Library maintainers	[OK]	Backend dependency injection	Keep container bindings consistent; a...
Logging (winston, morgan)	Technical	Library maintainers	[OK]	Operational visibility	Use Winston; structured logs; reduc...
Testing (Jest, Supertest, Playwright)	Technical	OSS maintainers	[OK]	Regression protection	Pin versions; unit tests; keep one E2E ...
Docker runtime (local) + Docker C...	External	Docker	[OK]	Reproducible environment	Pin Docker-compose.yml; healthchecks
SVG asset hosting (raw)	Technical	Project repo	[OK]	Serving analytics images	Pin CDN; cache headers; fal...
Academic timeline (committee)	Organizational	University/committee	-	Milestone gates for development	Blackboard plan from defense; buffer we...
Browser support	External	Browser vendors	-	Reader accessibility	Define baseline (e.g., last 2 Chrome/...